

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Performance Engineering, Systems for Data Science

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, C++, C, TypeScript, JavaScript, Java, SQL, Bash

LLM & Agents

LLM agents, agent orchestration, tool use, provider routing / fallback, output verification, evaluation, OpenAI / Anthropic / Groq APIs, Ollama, llama.cpp, MCP

RAG & Retrieval

RAG, RRF, cross-encoder reranking, MMR, sentence-transformers, FAISS, cell-level provenance

Deep Learning

PyTorch, JAX, HuggingFace Transformers, NumPy, autodiff, model training, normalizing flows, diffusion, model evaluation & benchmarking

Backend & Cloud

FastAPI, Node.js, REST APIs, async services, Docker, AWS, GCP, Azure, Git, GitHub Actions

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production models; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**, accelerating model iteration.
- Selected for a measured **AI-assisted development pilot** (GitHub Copilot), routing code migrations, tech-debt PRs, and security-finding triage through AI workflows; maintained a versioned `copilot-instructions.md` conventions file to keep generated code aligned with the codebase, an early hands-on take on applying AI across the SDLC.
- Engineered the multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Shipped an **open-source LLM feature** in production for a social app, **4-bit quantized Llama-7B via llama.cpp** on a single VM with per-persona prompts, async queued generation, rate limits, and a content-safety filter, built end-to-end with no ML mentor.
- Designed **20+ REST APIs** (Node.js, Fastify, PostgreSQL) and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics.

Projects

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage **agentic AI pipeline** that plans typed retrieval, reranks pages, extracts entities with deterministic parsers before LLM fallback, **verifies outputs**, and returns cell-level provenance, with provider routing and fallback across **OpenAI and Groq**.
- Hardened the service with **150+ automated tests**, an evaluation harness, and pipeline instrumentation; deployed as a live demo on **Google Cloud Run**.

AutoEval - Agentic Evaluation & Benchmarking

github.com/Deva-1903/AutoEval

Python, IR evaluation tooling, Git automation

- Built an **agent that improves an evaluation suite**, proposing queries and labels, scoring them against multiple systems, and accepting only updates that raise discriminative power on a holdout-heavy fitness objective.
- Added **anti-Goodharting guardrails** (schema checks, duplicate rejection, read-only holdout protection) and Git-backed auditable updates with report cards, directly relevant to building and maintaining **agentic AI benchmarks**.

cf-ai-research-scout - Edge-Native Agent Orchestration

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, Durable Objects, WorkflowEntrypoint, WebSockets

- Built an edge-native **agent** that orchestrates multi-step research, fetching web sources, generating digests, and streaming responses over **WebSocket** in real time.
- Designed **durable workflow orchestration** (WorkflowEntrypoint + Durable Objects) so multi-step agent tasks survive process restarts and tool failures, with state persisted server-side.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.